

Motor Imagery Classification Using TSK Fuzzy Inference Neural Networks

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Abstract— Electroencephalography (EEG) signals are often employed in modern BCI (brain-computer interface) systems. For patients with irreversible nerve injury, motor imagery BCIs enable them to control devices such as artificial limbs simply with brain activities (i.e. imagination) without physical movement. In practice, EEG signals are measured with multiple electrodes and often contaminated with various kinds of artifacts and noise.

In this study, a novel EEG motor imagery classification approach is developed based on discrete wavelet transform (DWT) and TSK (Tagaki–Sugeno–Kang) fuzzy inference neural networks. By using energy ratios in the frequency bands that are related with motor imagery activities, fewer features are needed for the classifier, which makes the real-time implementation of BCI systems more feasible. DWT not only provides multi-resolution analysis for non-stationary EEG signals, but also aids for noise and artifact reduction. Furthermore, the TSK fuzzy neural network is able to handle imprecise information with adaptive learning. The proposed algorithm is tested on EEG signals of PhysioNet database and compared with other methods. Satisfactory computer simulation results are obtained.

Keywords— *Motor imagery; Fuzzy neural networks; EEG; Discrete wavelet transform*

I. INTRODUCTION

It is reported that approximately 20 million people in the United States suffer from irreversible nerve damage [1]. In recent years, neuroprosthetic devices with Brain Computer Interface (BCI) have become a viable solution for paralysis patients. These hardware-software integrated systems can detect human brain activities and interpret (translate) them into commands to control artificial limbs or robot arms.

Electroencephalography (EEG) is commonly used to study brain cortical electrophysiology. It represents the voltage fluctuations resulting from ionic current that flows within the neurons of the brain ([2]). Brain activities such as sensory, cognitive and motor processing can result in changes of EEG; this phenomenon is termed event-related desynchronization (ERD) or event-related synchronization (ERS) [3]. ERD/ERS quantifies motor events through changes in mu (8-13Hz) and beta (13-30 Hz) rhythms of the alpha and beta frequency bands when motor activity is

performed or imagined.

In clinical studies, EEG signals are measured using the electrodes placed on the surface of a subject's scalp. Fig. 1 demonstrates the international 10/20 system, a standard system for electrode positioning with 21 electrodes [4]. High-density sensor placement systems such as the 10/10 international system, use 64 electrodes placed throughout the scalp and are able to provide higher resolution at the cost of computational complexity. These systems are named based on the relationship between the location of an electrode and the underlying area of cerebral cortex. For example, the “10” and “20” refer to the fact that the actual distances between adjacent electrodes are either 10% or 20% of the total front-back or right-left distance of the skull. To specify electrode positions, we use a letter to indicate the brain lobe and a number to identify the hemisphere location. More specifically, the letters F, T, C, P and O stand for frontal, temporal, “central”, parietal, and occipital lobes, respectively. Even numbers (2, 4, 6, 8) refer to electrode positions on the right hemisphere, whereas odd numbers (1, 3, 5, 7) refer to those on the left hemisphere. In EEG measurement, electrodes are usually placed at specific locations that can best capture the signals of interest. For ERS/ERD activities such as hand movement imagery, electrodes placed near somatosensory cortex (C3, Cz, and C4) are usually used to sense sensorimotor rhythm (SMR).

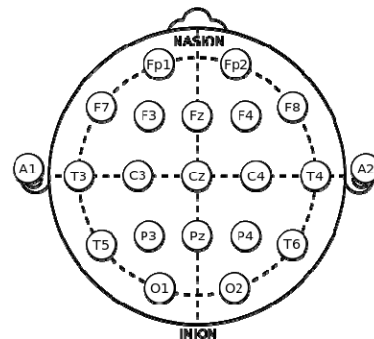


Fig. 1. The international 10/20 system for EEG sensor placement [4]

EEG signals are ideal for BCI applications due to their

low cost, ease of use, non-invasiveness, and the ability to extract information with high spatial-temporal resolution [5]. However, neuronal signals have to propagate through the Dura Mater and skull to get to the skin surface, and therefore have been more attenuated than other more invasive methods that use intracranial chronic electrodes. Furthermore, EEG signals are often contaminated with other signals and disturbances, such as electromyograms (EMG), electrooculargrams (EOG), power line, and environmental artifacts and have a very low SNR (signal to noise ratio) [6][7].

For EEG signal classification, the first step is to extract features from measurement. Various techniques have been proposed, including the Fast Fourier Transform (FFT), Discrete Wavelet Transform (DWT), Empirical Mode Decomposition (EMD), [8 - 11] etc. Recent studies show that, the multi-resolution property of DWT makes it a perfect tool for analyzing non-stationary physiological signals such as the EEG.

A reliable BCI system requires an accurate and efficient feature classification method. That is, the features extracted from measurement signals must be correctly classified to relate with the associated brain activities. Recently, the k-nearest neighbor (kNN) [12], support vector machines (SVM) [13], and multi-layer perceptron (MLP) [14] have been successfully applied for EEG signal discrimination.

Fuzzy logic is an extension of traditional Boolean logic. It is a multi-valued logic which is built upon the concept of fuzzy sets and membership functions [15]. Fuzzy systems attempt to reflect human reasoning and decision-making process and are able to handle vague and imprecise information. In recent years, with the emergence of artificial neural networks, studies on fuzzy neural (or neuro-fuzzy) systems have received more and more attention [16]. These systems combine the fuzzy “IF-THEN” inference with the adaptive learning ability of neural networks, and have been successfully applied to signal processing and control.

In this research, we propose a new approach for motor imagery classification using Takagi-Sugeno-Kang (TSK) fuzzy inference neural networks. Features that are related with imagined motor movements in theta, alpha, and beta frequency bands of EEG signals are extracted using discrete wavelet transform (DWT); and the TSK fuzzy neural network is employed to discriminate motor imageries. Unlike most wavelet-based feature extraction methods which use the approximation/detail coefficients directly, we use the normalized energy distribution in the associated frequency bands as features for classification. Computer simulations performed on PhysioNet dataset show that the proposed approach can dramatically reduce the number of features that are needed (and thus less system complexity and faster response which are more suitable for real-time implementation), while maintain satisfactory accuracy for motor imagery classification.

The rest of the paper is organized as follows. Section 2 first provides a summary on DWT and TSK fuzzy neural networks; then the proposed EEG classification approach is presented. Computer simulation results are discussed in

section 3, using data from PhysioNet database. Finally, section 4 concludes the paper and gives directions for future work.

II. THE PROPOSED APPROACH FOR EEG CLASSIFICATION

In this section, the proposed approach for EEG classification is discussed. First, a brief introduction on DWT and TSK fuzzy neural networks is presented in section 2.1 and 2.2; then the proposed approach is discussed in section 2.3.

2.1. The DWT (Discrete Wavelet Transform)

The continuous wavelet transform of a signal $x(t)$ is defined as [17]:

$$T(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} x(t) \psi^* \left(\frac{t-b}{a} \right) dt \quad (1)$$

where $\psi(\cdot)$ is called the “mother wavelet” and ψ^* represents its complex conjugate; a is the scaling factor and b is the shifting factor. In computer simulations, DWT (Discrete Wavelet Transform) is performed by selecting

$$a = 2^m, \quad b = 2^n \quad (2)$$

where m and n are positive integers.

Wavelet transform decomposes signals into multiple frequency bands. That is, wavelet transform can be considered as passing a signal through a set of low-pass (LP) and high-pass filters (HP). Through this process, a signal can be decomposed into different levels. At each level, it contains a set of detail coefficients (D) and approximation coefficients (A), as shown in Fig. 2, where “ $\downarrow 2$ ” represents down sampling with a factor of 2.

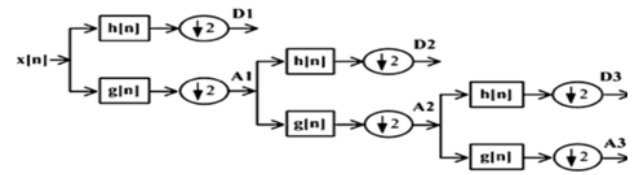


Fig. 2. The Discrete Wavelet Transform (DWT) [18]

2.2. The TSK (Tagaki-Sugeno-Kang) Fuzzy Neural Networks

A typical TSK fuzzy neural network with two inputs and one output is illustrated in Fig. 3 [15]. It can implement the following four fuzzy “if-then” rules of Takagi-Sugeno’s type:

- 1: x_1 is A_1 and x_2 is B_1 , then y is f_1
- 2: x_1 is A_2 and x_2 is B_2 , then y is f_2
- 3: x_1 is A_2 and x_2 is B_1 , then y is f_3
- 4: x_1 is A_1 and x_2 is B_2 , then y is f_4

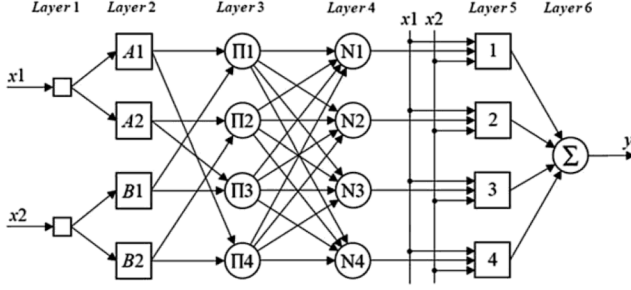


Fig. 3. Takagi-Sugeno-Kang Style Inference Neural Network [15]

Layer 1 simply takes and distributes the inputs. Layer 2 is the membership (also called fuzzification) layer, where the degree of membership is calculated for each input. The membership function is often in bell shape; and Gaussian function is the most commonly used one:

$$y_{Ai} = A_i(x_1) = \exp\left[-\frac{(x_1 - m_{Ai})^2}{\sigma_{Ai}^2}\right] \quad (3)$$

where $i = 1, 2$; m_{Ai} and σ_{Ai} is the center and width of membership function, respectively. Similar relationship also applies to node B1 and B2 for input x_2 , respectively. Parameters in this layer (e.g. m_{Ai} and σ_{Ai}) are often referred as “premise parameters”.

Layer 3 is called rule layer. The output of node Π_i in Fig. 3 can be expressed as ($i = 1, 2, 3, 4$):

$$y_{\Pi i} = y_{Ai} \times y_{Bi} \triangleq w_i \quad (4)$$

Layer 4 is the normalization layer; outputs of this layer are also called “normalized firing strengths”. The output of node N_i ($i = 1, 2, 3, 4$) can be expressed as:

$$y_{Ni} = \frac{y_{\Pi i}}{\sum_k w_k y_{\Pi k}} \triangleq \bar{w}_i \quad (5)$$

where k is the index of outputs from the previous layer (in Fig. 3, $k = 1, 2, 3, 4$).

Layer 5 (termed fuzzy inference layer) performs defuzzification operation. Note each node in this layer not only receives the output from the previous layer, but also includes the inputs. Let y_i ($i = 1, 2, 3, 4$) be the output of the i -th node in layer 5, we have:

$$y_i = y_{Ni} \cdot f_i(x_1, x_2) = \bar{w}_i f_i(x_1, x_2) \quad (6)$$

For TSK network, $f_i(\cdot)$ are defined as:

$$f_i(x_1, x_2) = p_i x_1 + q_i x_2 + r_i \quad (7)$$

$$\begin{aligned} y_i &= \bar{w}_i (p_i x_1 + q_i x_2 + r_i) \\ &= (\bar{w}_i x_1) p_i + (\bar{w}_i x_2) q_i + (\bar{w}_i) r_i \\ &= [\bar{w}_i x_1 \quad \bar{w}_i x_2 \quad \bar{w}_i] \begin{bmatrix} p_i \\ q_i \\ r_i \end{bmatrix} \end{aligned} \quad (8)$$

Parameters in this layer (e.g. p_i, q_i, r_i) are often referred as

“consequent parameters”. In matrix form, the above equation can be written as

$$AX = B \quad (9)$$

where X is the vector of all consequent parameters; each element of B vector is y_i ; and each row of A matrix is $[\bar{w}_i x_1 \quad \bar{w}_i x_2 \quad \bar{w}_i]$, $i = 1, 2, 3, 4$.

Layer 6 generates output by combining outputs (with index i) from the previous layer:

$$y = \sum_i y_i = \sum_i \bar{w}_i f_i = \frac{\sum_i w_i f_i}{\sum_i w_i} \quad (10)$$

In this study, a hybrid learning algorithm is employed to train the TSK neural network [15] [16]. That is, we use Back Propagation (BP) algorithm to tune the premise parameters and Least Squares Estimate (LSE) to find the consequent parameters. Let α be the vector of all premise parameters, it is updated as:

$$\alpha_{n+1} = \alpha_n - \eta \frac{\partial J}{\partial \alpha} \quad (11)$$

where n is the index of iteration; J is the objective function to be minimized; η is the learning rate. Recall that X is the vector of all consequent parameters, we have:

$$S_{i+1} = \frac{1}{\lambda} \left[S_i - \frac{S_i a_{i+1}^T a_{i+1}^T S_i}{\lambda + a_{i+1}^T S_n a_{i+1}} \right] \quad (12)$$

$$X_{i+1} = X_i + S_{i+1} a_{i+1} (b_{i+1}^T - a_{i+1}^T X_i) \quad (13)$$

where $i = 0, 1, 2, \dots, P-1$ (P is the total number of training pairs); λ is the forgetting factor. S is the covariance matrix of the coefficients of the consequent parameters (see Eq. (9)); a_i is the i -th row of the A matrix defined below Eq. (9) and b_i is the i -th row of B .

2.3. The Proposed Approach

The flow chart of the proposed approach for EEG classification is shown in Fig. 4.

In wavelet-based feature extraction methods, the approximation and/or detail coefficients are often directly used as features. However, this may result in excess number of features and even trigger the issue of “the curse of dimensionality”. In [19], an approach based on energy distribution is proposed for seizure detection. In this research, we adopt this method for motor imagery classification of EEG signals. By using the normalized energy distribution in the associated frequency bands, the number of features can be dramatically reduced and thus resulting less system complexity and faster response which are more suitable for real-time implementation.

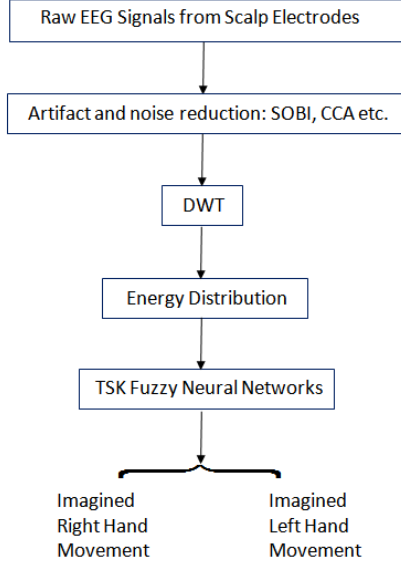


Fig. 4. The computational flow chart for neural network training

Let E_T be the total energy of the signal, A_i be the approximation coefficients and D_i be the detail coefficients ($i = 1, 2, \dots, L_l$, where L_l is the total number of approximation or detail coefficients of the first level decomposition), then:

$$E_T = \sum_{i=1}^{L_l} (A_i)^2 + \sum_{i=1}^{L_l} (D_i)^2 \quad (14)$$

Similarly, let C to be the approximation or detail coefficients of the specific decomposition level in interest (m). If it contains L_m coefficients, the specific energy E_s can be written as:

$$E_s = \sum_{i=1}^{L_m} (C_i)^2 \quad (15)$$

Finally, the energy ratio E_r , represented by percentage, can be calculated by:

$$E_r = \frac{E_s}{E_T} \times 100 (\%) \quad (16)$$

III. SIMULATION RESULTS

In this section, the performances of the proposed approach is studied and compared with other methods via computer simulation.

The EEG signals used in this research come from the Motor Movement and Imagery (MMI) Database available online at PhysioNet [20]. The data consists of more than 1,500 EEG recordings acquired from 109 subjects recorded at 160 Hz using BCI2000 with 64-channel EEG for a total of 14 experimental runs. BCI2000 is a general-purpose software that has been widely used for data acquisition, stimulus presentation, and brain monitoring applications in

BCI research [21].

In this research, we consider the classification of two imagined motor movements: open/close left fist vs. open/close right fist. When the experiment starts, baseline assessment (T0) is performed one time per test subject for a one-minute period; then a target appears on either the left or right side of the screen (annotated as T1 or T2, respectively), which prompts the subject to open and close the corresponding fist until the target disappears after approximately 4.1 seconds. Each trial is followed by 4.1 seconds of rest [22]. Overall, the left and right fist activity was engaged in a total of 15 trials for three separate periods.

Raw EEG recordings are often contaminated by various kinds of artifacts and noise. Sweating and drift in electrode impedance may gradually change the voltage measurement, leading to saturation in amplitude and distortions over event-related time course [23]. Muscle contraction artifacts, such as electrooculographic (EOG) and electromyographic (EMG), may seriously degrade the quality of EEG signals due to their relative high amplitude. In general, EEG signal amplitudes have an average of 50 μ V magnitude with a maximum of 100 μ V; while the amplitude of an EOG signal is generally in the range between 50 to 200 μ V. For EMG signals, the range of amplitude is usually between 20 to 200 μ V and can reach as high as 1.5 mV [23].

In this research, EEG signals are pre-processed (before feature extraction) to reduce the effects of artifacts and noise. Power line disturbance at 60 Hz or 50 Hz can be eliminated by a notch filter. However, artifacts such as EOG and EMG are difficult to remove due to the spectrum overlap with brain neuron activities.

Blind source separation (BSS) is the process that separates signals of different sources from a given mixed signal. It is based on the assumption that source signals are statistically independent and superimposed in a linear manner. For EEG signal, we assume that the signal recorded from scalp is the spatially stable mixture of the activities of temporally independent cerebral and artifactual sources, and summation of potentials arising from different parts of the brain, scalp, and body is linear at the electrodes [24] [25]. Define a set of individual source signals:

$$s(t) = [s_{EEG}(t) \quad s_{EEG}(t) \quad s_{EOG}(t)]^T \quad (17)$$

The mixed signal (EEG measurement) can then be written as:

$$x(t) = M s(t) \quad (18)$$

where M is called matrix of transfer coefficients. To remove the effect of EOG, Second Order Blind Identification (SOBI) is employed. It uses the second order statistics to find spatial components that have non-zero autocorrelations and zero cross-correlations (both correlations are defined in time domain). Canonical Correlation Analysis (CCA) is mainly used for the removal of EMG. In CCA, EEG signal is considered as a combination of maximally autocorrelated

and mutually uncorrelated spatial components; spatial components with the lowest autocorrelation values are assumed to be related to artifacts and thus removed.

To capture the grasping motion of the hand, Event Related Potentials (ERPs) recorded from electrodes C3, Cz, and C4, both placed above the motor cortex, are considered in this study. Spectral analysis is performed with DWT. To detect brain activities in mu and beta rhythms, EEG signals are decomposed to six levels, using Debauchee wavelet with two vanishing moments. Instead of using DWT coefficients directly, or the statistic characteristics of the coefficients (e.g. RMS (root-mean-square value), MAV (mean absolute value), etc.), we consider the event associated energy pattern in certain frequency bands. For the energy change in EEG signals caused by brain neuron activities on motor imagery, delta, theta, alpha, and beta frequency bands should be included. Thus, we choose detail coefficients D4, D5, and D6, and use Eq. (14) – (16) to find the normalized energy (in percentage) in each frequency band. These features then become the input vector of TSK fuzzy neural network.

As discussed above, the proposed TSK fuzzy neural network has nine inputs which represent the normalized energy in three different frequency bands from three electrodes. Gaussian function is selected for the membership function, as in Eq. (6). To train the Fuzzy neural network, all the data are partitioned into three mutually exclusive sets: training (70%), validation (15%), and testing (15%). A hybrid training algorithm (Eq. (11) – (13)) which combines the Back Propagation (to tune the premise parameters) and the Least Squares Estimate (to tune the consequent parameters) is employed. The fuzzy inference network has a single output with binary values to discriminate the imaged left/right hand movement.

To fully evaluate the performance of the proposed approach, four different statistical measures are considered, i.e. specificity (true negative rate), sensitivity (true positive rate), selectivity (precision), and accuracy [26]:

$$\text{Specificity} = \text{TN}/(\text{TN} + \text{FP}) \quad (19)$$

$$\text{Sensitivity} = \text{TP}/(\text{TP} + \text{FN}) \quad (20)$$

$$\text{Selectivity} = \text{TP}/(\text{TP} + \text{FP}) \quad (21)$$

$$\text{Accuracy} = (\text{TN} + \text{TP})/(\text{TN} + \text{TP} + \text{FN} + \text{FP}) \quad (22)$$

where TP represents “true positive”, TN is for “true negative”, FN represents “false negative”, and FP is for “false positive”, respectively. In other words, TP and TN represent the results of correct classification, while FP and FN represent the results of incorrect classification. Note that high accuracy means overall better classification results for both classes, and thus has been widely used in other research papers. Table 1 summarizes our test results, averaged with 40 subjects.

Table 1. Performance of the Proposed Approach (%)

Specificity	Sensitivity	Selectivity	Accuracy
79.02	79.90	79.66	79.44

The proposed approach is also compared with other methods, including Empirical Mode Decomposition [8], DWT & Support Vector Machine [13]. Table 2 shows that the accuracy of our proposed approach is comparable with other techniques, with an improvement on mean accuracy of 1.7% to 4.4%, respectively. It should be noted that our proposed approach uses fewer EEG channels (three vs. 11 used in [8] and 6 used in [13]), which makes it more suitable for real-time implementation. In addition, our test results are averaged on 40 subjects while only 10 subjects are considered in [8]. Besides, instead of using DWT coefficients directly, we use the normalized energy in certain frequency bands which can dramatically reduce the dimension of feature vector and speed up the training process.

Table 2. Comparison with Other Methods

	Mean Accuracy (%)	Max. Accuracy (%)
Multivariate Empirical Mode Decomposition [8]	77.7	97.4
DWT & Support Vector Machine [13]	75.0	N/A
The Proposed Approach DWT & TSK Fuzzy NN	79.4	95.0

IV. CONCLUSIONS AND FUTURE WORK

EEG is a non-invasive measurement of physiological signals using the electrodes placed on the surface of a subject’s scalp. In this study, a novel EEG motor imagery classification approach is developed based on discrete wavelet transform and TSK (Tagaki–Sugeno–Kang) fuzzy inference neural networks. The proposed algorithm uses energy ratio (i.e. normalized energy distribution) in certain frequency bands to detect and classify motor imagery activities. Computation simulation results using EEG signals of PhysioNet database show that the performance of our approach is comparable with other techniques with 1.7% to 4.4% improvement on mean accuracy, and also has the advantage of using fewer EEG channels which can release the computation burden and makes it more suitable for real-time implementation.

For future work, we plan to perform more tests to fully study the performance of the proposed approach under different SNRs (signal to noise ratio).

REFERENCES

- [1] D. Grinsell and C. P. Keating, "Peripheral Nerve Reconstruction after Injury: A Review of Clinical and Experimental Therapies", BioMed Research International, vol. 2014.
- [2] Electroencephalography, <https://en.wikipedia.org/wiki/Electroencephalography>
- [3] G.Pfurtscheller, "Functional brain imaging based on ERD/ERS", Vision Research, Volume 41, Issues 10–11, May 2001, p. 1257-1260.
- [4] 10–20 system (EEG), [https://en.wikipedia.org/wiki/10%E2%80%9320_system_\(EEG\)](https://en.wikipedia.org/wiki/10%E2%80%9320_system_(EEG))
- [5] Burle, Boris, Laure Spieser, Clémence Roger, Laurence Casini, Thierry Hasbroucq, and Franck Vidal. "Spatial and temporal resolutions of EEG: Is it really black and white? A scalp current density view." International Journal of Psychophysiology 97, no. 3 (2015): 210-20.
- [6] Gomez-Herrero, German, Wim Clercq, Haroon Anwar, Olga Kara, Karen Egiazarian, Sabine Huffel, and Wim Paesschen. "Automatic Removal of Ocular Artifacts in the EEG without an EOG Reference Channel." *Proceedings of the 7th Nordic Signal Processing Symposium - NORSIG 2006* (2006).
- [7] Clercq, Wim De, A. Vergult, B. Vanrumste, W. Van Paesschen, and S. Van Huffel. "Canonical Correlation Analysis Applied to Remove Muscle Artifacts From the Electroencephalogram." *IEEE Transactions on Biomedical Engineering* IEEE Trans. Biomed. Eng. 53.12 (2006): 2583-587.
- [8] Park, Cheolsoo, David Looney, Naveed Ur Rehman, Alireza Ahrabian, and Danilo P. Mandic. "Classification of Motor Imagery BCI Using Multivariate Empirical Mode Decomposition." *IEEE Trans. Neural Syst. Rehabil. Eng.* IEEE Transactions on Neural Systems and Rehabilitation Engineering 21.1 (2013): 10-22.
- [9] Zhang, Jin, Guoli Li, Chao Xu, Deyang Shao, and Long Jiao. "Study of Harmonic Analysis Based on Improved Discrete Fourier Transform." *2016 IEEE 11th Conference on Industrial Electronics and Applications (ICIEA)* (2016).
- [10] H. Yuan, A. Doud, A. Gururajan, and B. He, "Cortical imaging of event-related (de)synchronization during online control of brain-computer interface using minimum-norm estimates in frequency domain," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 16, no. 5, pp. 425–431, Oct. 2008.
- [11] Alomari, Mohammad H., Emad A. Awada, Aya Samaha, and Khaled Alkamha. "Wavelet-Based Feature Extraction for the Analysis of EEG Signals Associated with Imagined Fists and Feet Movements." *CIS Computer and Information Science* 7.2 (2014).
- [12] Nurul E'zzati Md Isa, Amiza Amir, Mohd Zaizu Ilyas and Mohammad Shahrzazel Razalli, "The Performance Analysis of K-Nearest Neighbors (K-NN) Algorithm for Motor Imagery Classification Based on EEG Signal", 2017 International Conference on Emerging Electronic Solutions for IoT (ICEESI 2017).
- [13] Hajibabazadeh, Mahdiyeh, and Vahid Azimirad. "Brain-robot Interface: Distinguishing Left and Right Hand EEG Signals through SVM." 2014 Second RSI/ISM International Conference on Robotics and Mechatronics (ICRoM) (2014).
- [14] Fei-yan Fan, Ying-jie Li, Yi-hong Qiu, Yi-sheng Zhu, "Use of ANN and Complexity Measures in Cognitive EEG Discrimination", 27th IEEE-EMBS Annual International Conference of the Engineering in Medicine and Biology Society, 2016.
- [15] Rafik Aziz Aliev, Babek Ghalib Guirimov, "Type-2 Fuzzy Neural Networks and Their Applications", Springer, 2014.
- [16] Jang, J.-S.r. "ANFIS: Adaptive-network-based Fuzzy Inference System." *IEEE Transactions on Systems, Man, and Cybernetics* IEEE Trans. Syst., Man, Cybern. 23.3 (1993): 665-85.
- [17] Addison, Paul S. "Wavelet transforms and the ECG: a review." *Physiological Measurement* 26, no. 5 (2005).
- [18] Discrete wavelet transform, http://en.wikipedia.org/wiki/Discrete_wavelet_transform
- [19] Omerhodzic, I., Avdakovic, S., Nuhanovic, A., Dizdarevic, K. "Energy Distribution of EEG Signals: EEG Signal Wavelet-Neural Network Classifier". World Academy of Science, Engineering and Technology, (2010) , pp 1066–1071.
- [20] EEG Motor Movement/Imagery Dataset, <http://physionet.org/pn4/eegmmidb/>.
- [21] Schalk, G., D.j. Mcfarland, T. Hinterberger, N. Birbaumer, and J.r. Wolpaw. "BCI2000: A General-Purpose Brain-Computer Interface (BCI) System." *IEEE Transactions on Biomedical Engineering* IEEE Trans. Biomed. Eng. 51.6 (2004).
- [22] Goldberger AL, Amaral LAN, Glass L, Hausdorff JM, Ivanov PCh, Mark RG, Mietus JE, Moody GB, Peng C-K, Stanley HE. PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals. *Circulation* 101(23):e215-e220.
- [23] Kappenman, Emily S., and Steven J. Luck. "The effects of electrode impedance on data quality and statistical significance in ERP recordings." *Psychophysiology*, 2010.
- [24] Shayegh, Farzaneh, and Abbas Erfanian. "Real-Time Ocular Artifacts Suppression from EEG Signals Using an Unsupervised Adaptive Blind Source Separation." *2006 International Conference of the IEEE Engineering in Medicine and Biology Society*, 2006.
- [25] Hamaneh, Mehdi Bagheri, Numthip Chitravas, Kitti Kaiboriboon, Samden D. Lhatoo, and Kenneth A. Loparo. "Automated Removal of EKG Artifact From EEG Data Using Independent Component Analysis and Continuous Wavelet Transformation." *IEEE Transactions on Biomedical Engineering* 61, no. 6 (2014): 1634-641.
- [26] Powers, David M W. "Evaluation: From Precision, Recall and F-Measure to ROC, Informedness, Markedness & Correlation". *Journal of Machine Learning Technologies* (2011). 2 (1) : 37–63.